Extracting the data from the Smyth et al. 1969 paper for building a drug-drug interaction LD50 dataset

Notebook link - https://github.com/summyyyyy/Mixture-Toxicity-Predictor-with-Real-Time-Explanations/blob/Dataset-Preparation/Data_extraction_from_the_Smyth_et_al_1969_paper_for_building_a_DDI_LD50_dataset.ipynb

Reference - SMYTH, H.F. et al., 1969. An exploration of joint toxic action: Twenty-seven industrial chemicals intubated in rats in all possible pairs. *Toxicology and Applied Pharmacology*, 14(2), pp. 340–347. Available from: https://www.sciencedirect.com/science/article/pii/0041008X6990115X

1. Locate the list of chemicals

The 27 industrial chemicals tested are listed in the Materials section on page 2. The list appears under the heading MATERIALS and includes names such as Acetone, Acetonitrile, Acrylonitrile, and others. These names become the row and column labels for the interaction data.

2. Find the single drug LD50 values

Table 3 on page 7 provides the median LD50 of each chemical when tested alone. The table has two columns. The first column lists the chemical names. The second column gives the LD50 value in milliliters per kilogram (ml/kg). These values are needed to calculate the predicted additive toxicity for each mixture.

3. Extract the ratio table

Table 1 on page 3 contains the core interaction data. It is a large triangular table showing the ratio of predicted LD50 to observed LD50 for every pair of chemicals. The chemicals are arranged in the same order as in the materials list. The numbers in the table represent the ratio for the pair formed by the chemical in that row and the chemical in that column. A ratio below 1.00 means the mixture was more toxic than predicted. A ratio above 1.00 means the mixture was less toxic than predicted.

4. Understand the table layout

Table 1 is printed as a lower triangular matrix. The chemical names run down the left side and across the top in the same order. The diagonal is empty because a chemical is not paired with itself. For each cell the number corresponds to the pair of the row chemical and the column chemical. To build a complete dataset each unique combination of two different chemicals must be recorded along with its ratio.

5. Record each pair and its ratio

Going through the table row by row for each chemical every value in that row is matched with the corresponding column chemical. This creates a list of all 350 pairs. The pair order for example Acetone with Acetonitrile and Acetone with Acetophenone is recorded. The ratio value from the table is noted for each pair. Some cells may be empty because the pair was not tested. The paper mentions that one pair Morpholine with Ethyl acrylate reacted violently and could not be tested. These missing entries should be skipped.

6. Identify the deviant pairs

Table 2 on page 6 lists the pairs that deviated most from additive action. These are the pairs whose ratios fall outside the range of 0.40 to 2.70. The table separates them into two groups. One group is less than additive meaning the mixture was more toxic than predicted with ratios below 0.40. The other group is more than additive meaning the mixture was less toxic than predicted with ratios above 2.70. These pairs are valuable for testing model performance on non additive interactions.

7. Collect SMILES strings for each chemical

SMILES strings are not provided in the paper. They must be obtained from external sources such as PubChem or ChemSpider. Each chemical name from the list is searched in a chemical database. The canonical SMILES string for that compound is recorded. For proprietary substances like Tergitol XD and Ucon fluids exact SMILES may not be available. Approximate representations based on general composition can be used with a note in the documentation.

8. Calculate the mixture LD50

The mixture LD50 is not directly given in the paper. It can be calculated from the ratio and the predicted additive LD50. The predicted additive LD50 for a 1 to 1 by volume mixture is obtained from the harmonic mean formula. The formula is one over predicted LD50 equals one half divided by LD50 of chemical A plus one half divided by LD50 of chemical B. This gives the predicted additive LD50. The observed mixture LD50 is then the predicted additive LD50 divided by the ratio from Table 1. This calculation produces a continuous LD50 value for each pair.

9. Organize the data into a table

All collected information is assembled into a structured table. The table should have one row for each unique drug pair. The columns include drug A name drug B name SMILES for drug A SMILES for drug B single drug LD50 for drug A single drug LD50 for drug B mixture LD50 and the unit which is milliliters per kilogram. The ratio from Table 1 can also be included as an additional column if needed for reference.

10. Verify the extracted data

The extracted data should be checked against the original paper to ensure accuracy. A few sample pairs can be verified by recalculating the mixture LD50 from the reported ratio and the individual LD50 values. This confirms that the extraction and calculations were performed correctly.